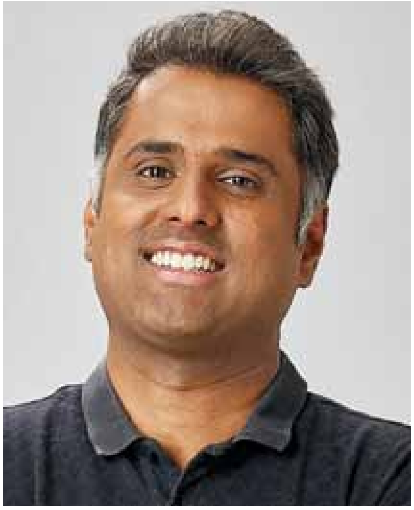
four kilometres per hour. The entire identification, target selection, and precision spraying process must occur within a brief timeframe of just 100 milliseconds," elaborates Rao.

The startup manufactures the carrier board for the system-on-chip (SoC). It incorporates the FPGA, various components, and the required power infrastructure to make the camera system resilient in temperatures ranging from 50 to 70 degrees Celsius, which are commonly found in outdoor agricultural environments.

Rao explained the reason for constructing their carrier board and the challenges that it resolved. "Previously, we used a master-slave unit with two cameras sharing one SoC. This could lead to the failure of both cameras if the main unit failed. The camera needs vary, based on equipment specifications. We tried a shared AI resource concept last season, but it wasn't optimal, leading to



Jaisimha Rao, Founder

the latest design's development," he elucidates.

The company has launched 50 robots and is targeting to spray an area of over one hundred thousand acres in India. The robots have been deployed in five cities—Akola (16), Guntur (16), Khammam (9), Bellary (7), and Mettur (2). The company offers its technology to village-level entrepreneurs who lease the equipment and provide spraying services to farmers on a per-acre basis. "These individuals, both men and women, are deeply connected with the land, overseeing around 502,000 acres of catchment area. Many of them are already involved in activities like pesticide or tractor dealerships. They entered into a leasing arrangement with us for a year, gaining access to the machine," Rao says.

The startup has raised about ₹700 million in funding over three and a half years and is planning to raise more. With a team of around 53 full-time employees, additional interns and field-level officers, the company plans to expand beyond India, targeting markets in North America.

—*Yashasvini Razdan*

2 THIS STARTUP'S 2KW BATTERIES CAN BE CHARGED 20% TO 80% IN 20 MINUTES

EMO Energy claims that 20 minutes of charging helps its 2kWh batteries provide up to 50km of charge to E2Ws and E3Ws. It is utilising a digital 4.0 setup, including robots and a connected assembly line, for manufacturing batteries

Mrinal Sinha, Co-Founder at EMO Energy, in a conversation with Electronics For You, shared that the 2kW batteries his startup has developed can be charged from 20% to 80% within 20 minutes using a standard home socket.

"The only limitation is the charger you are using. It has to be a 3.3kW charger. Although we can supply these chargers to original equipment manufacturers (OEMs), we do not restrict them from sourcing them from a third party," he says.